

---

## Lecture 5

### Exercise 2

Q2: Integrate the expression  $f(x) = \sin(x) e^{-x}$ , and then take its derivative.

```
In[7]:= f[x_] := Sin[x] Exp[-x];

F[x_] = Integrate[f[x], x];

Fp[x_] = Simplify[D[F[x], x]];

Print["Function: f(x) = ", f[x]]

Print["Primitive function: F(x) = ", F[x]]

Print["Derivative of the primitive: F'(x) = ", Fp[x]]
```

Function:  $f(x) = e^{-x} \sin(x)$

Primitive function:  $F(x) = -\frac{1}{2} e^{-x} (\cos(x) + \sin(x))$

Derivative of the primitive:  $F'(x) = e^{-x} \sin(x)$

### Exercise 3

Q3: Solve the equation  $x \ln(x) - 3x + 10 = 6$ , both symbolically and numerically.

```
In[13]:= eq = x Log[x] - 3 x + 10 == 6;

Print["Equation: ", eq]

(*Numeric solution*)
xNum = Sort[x /. NSolve[eq, x, Reals]];

Print["Numeric solutions = ", xNum]

(*Symbolic solution*)
sym = Reduce[eq, x, Reals];

Print["Symbolic solution: ", sym]

(*Numeric values of the symbolic solutions*)
xSym = Cases[sym, x == val_ -> val, Infinity];

xSymNum = Sort[N[xSym]];

Print["Numeric values of symbolic solutions = ", xSymNum]

(*Comparison of results*)
Print["Symbolic and numeric solution matching: ", Sort[xSymNum] === Sort[xNum]]
```

Equation:  $10 - 3x + x \log[x] == 6$

Numeric solutions = {1.56883, 15.5229}

Symbolic solution:  $x == e^{3 + \text{ProductLog}\left[-\frac{4}{e^3}\right]}$  ||  $x == e^{3 + \text{ProductLog}\left[-1, -\frac{4}{e^3}\right]}$

Numeric values of symbolic solutions = {1.56883, 15.5229}

Simbolic and numeric solution matching: True